ACCIDENTXAI

Temporal Explainable AI for Accident Severity Prediction

Research Report

Dataset: US Accidents Dataset (March 2023)

Records: 7.7 Million Accident Records

Time Period: 2016-2023

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Executive Summary

This research project presents a comprehensive explainable AI (XAI) pipeline for predicting accident severity using temporal SHAP analysis on a large-scale US traffic accident dataset spanning 2016-2023.

Key Findings:

- LightGBM achieved the best performance with F1-Weighted score of 0.8744 and ROC-AUC of 0.6984
- Year emerged as the dominant predictor, with SHAP importance of 1.38, showing dramatic drift after 2020
- Temporal analysis revealed significant feature importance shifts during the pandemic period (2020), particularly for Month and Night_Driving
- Geographic features (Start_Lat, Start_Lng) and temporal patterns (Weekday, Rush_Hour) consistently influenced predictions across all years
- Infrastructure features showed varying stability, with Traffic_Signal importance declining over time

1. Introduction

Research Objectives

This study investigates accident severity prediction through the lens of explainable AI, with emphasis on understanding how feature importance evolves over time. The primary objectives include:

- Predicting accident severity using multiple machine learning models
- Analyzing temporal drift in feature importance across 2016-2023
- Identifying stable versus unstable accident risk factors
- Providing interpretable insights through SHAP-based analysis

Dataset Description

The US Accidents Dataset (March 2023 version) comprises approximately 7.7 million accident records collected from 2016 to 2023. The dataset includes comprehensive information about accident characteristics:

- Environmental features: temperature, humidity, pressure, visibility, wind speed, weather conditions
- Road infrastructure indicators: amenities, bumps, crossings, junctions, signals, etc.
- Temporal attributes: year, month, day, hour, weekday, rush hour, night driving, sunrise/sunset
- Geospatial features: latitude, longitude, state location

Challenges and Motivation

The research addresses critical challenges in accident severity prediction:

- Severe class imbalance in the target variable

- Temporal leakage prevention in train-test splitting
- Understanding evolving accident patterns over time
- Providing explainable predictions beyond black-box model outputs

2. Methodology

Data Preprocessing

The preprocessing pipeline included:

- Missing value handling and imputation
- Categorical feature encoding and cleaning
- Feature engineering: Rush_Hour, Night_Driving, temporal components
- Binary target construction for severity classification

Train-Test Strategy

To prevent temporal leakage, we employed temporal train-test splitting instead of random splitting. The dataset was partitioned chronologically, ensuring that training data never contains information from future time periods relative to test data. Class imbalance was addressed using SMOTE (Synthetic Minority Over-sampling Technique) applied exclusively to training data.

Model Selection

Five models were compared:

- Baseline Models: Logistic Regression, Random Forest
- Boosting Models: XGBoost, LightGBM, CatBoost

Explainability Analysis

SHAP (SHapley Additive exPlanations) values were computed to interpret model predictions. Analysis included:

- Global feature importance ranking
- SHAP summary plots showing feature-output relationships
- Year-by-year SHAP importance tracking (2016-2023)
- Feature drift analysis quantifying importance changes over time

3. Results and Analysis

Model Performance Comparison

Table 1 presents performance metrics for all five models:

Model	F1 Macro	F1 Weighted	Precision	Recall	ROC-AUC
LightGBM	0.5259	0.8744	0.3038	0.0615	0.6984
XGBoost	0.5169	0.8731	0.2874	0.0491	0.6945
CatBoost	0.4841	0.8701	0.3333	0.0076	0.6934
Random Forest	0.5177	0.8742	0.3286	0.0484	0.6762
Logistic Regression	0.4423	0.6742	0.0978	0.4481	0.5486

Table 1: Model performance comparison across evaluation metrics. LightGBM achieved the highest F1-Weighted (0.8744) and ROC-AUC (0.6984) scores.

Key Observations:

- Ensemble boosting models consistently outperformed baseline models
- LightGBM achieved best overall performance
- Logistic Regression showed higher recall but poor precision
- Severe class imbalance limited recall performance across all models

Global Feature Importance

SHAP analysis identified the following top 10 most important features for the best-performing LightGBM model:

Feature	Mean SHAP Importance
Year	1.3804
Weekday	0.2553
Start_Lat	0.2483
Rush_Hour	0.2393
Start_Lng	0.2071
Hour	0.2002
State	0.1628
Crossing	0.1265
Month	0.1262

Traffic_Signal	0.1014

Table 2: Top 10 features ranked by mean SHAP importance. Year dominates predictions with importance of 1.38, followed by temporal and geographic features.

The SHAP summary plot (Figure 1) visualizes how individual feature values impact predictions:

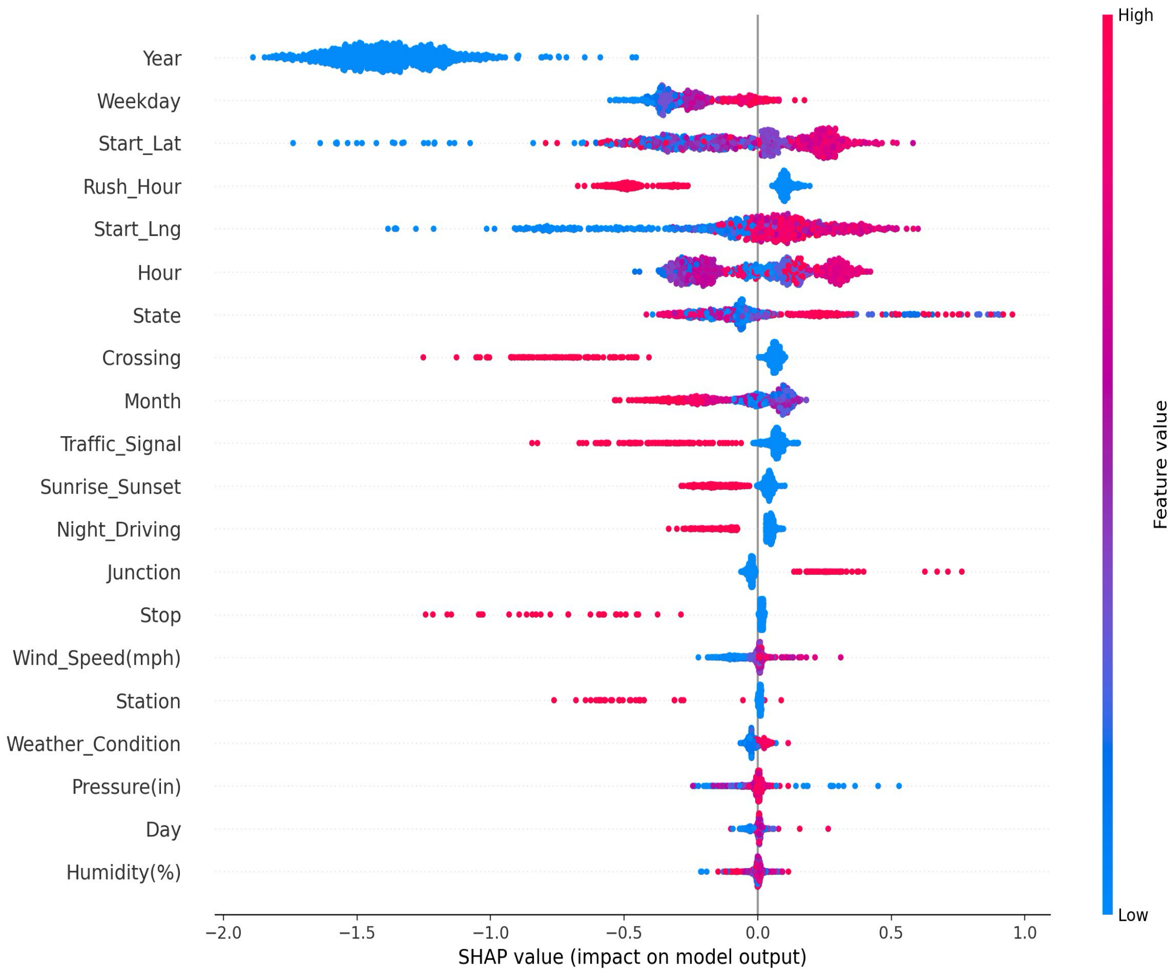

Figure 1: SHAP summary plot showing feature contributions to model output. Blue points indicate low feature values with negative impact, red points show high values with positive impact.

Temporal SHAP Drift Analysis

Year-by-year analysis of feature importance revealed significant temporal dynamics:

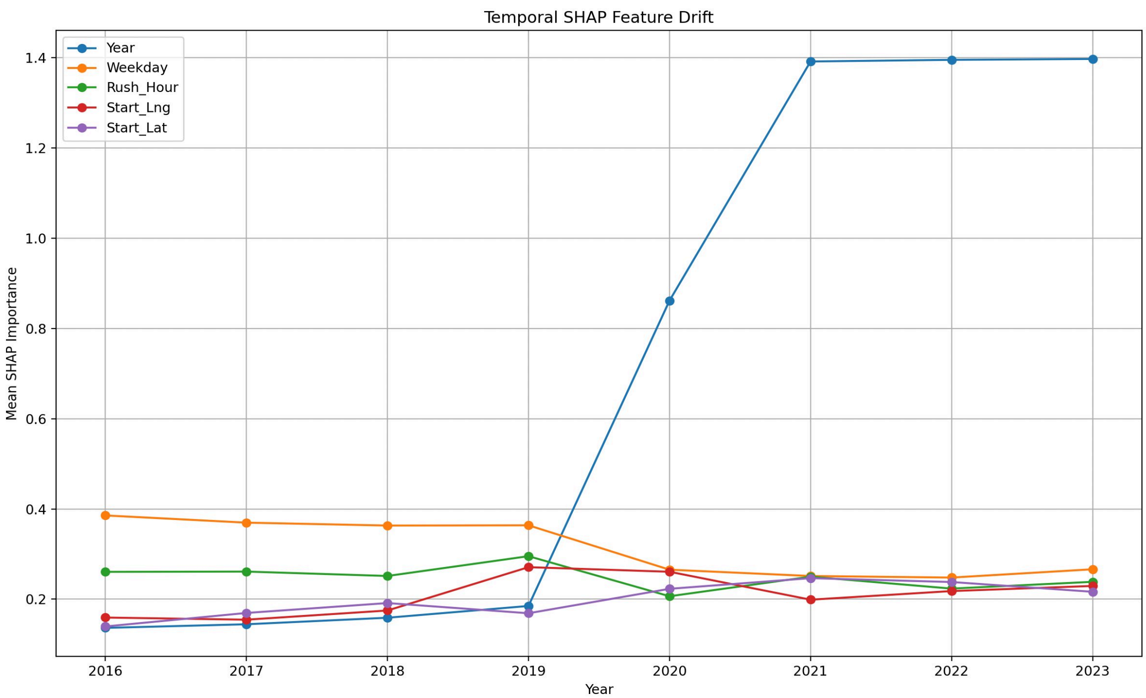

Figure 2: Temporal SHAP feature drift from 2016 to 2023. Year shows dramatic increase after 2020, while Weekday and Rush_Hour remain relatively stable.

Major Temporal Findings:

- Year feature shows the largest drift range (1.26), indicating strong temporal distribution shift
- Month importance spiked dramatically in 2020 (0.339), suggesting pandemic-related seasonality changes
- Night_Driving importance increased substantially during pandemic years (0.068-0.135)
- Traffic_Signal importance declined over time, potentially reflecting infrastructure normalization
- Weekday and Rush_Hour remained consistently important throughout the period

Feature Importance Heatmap

The temporal heatmap provides a comprehensive view of feature importance evolution:

Figure 3: Feature importance heatmap across all years. Darker colors indicate higher importance. Month's spike in 2020 is clearly visible.

Feature Drift Analysis

Quantitative drift analysis measured importance variation across all years:

Feature	Min	Max	Mean	Drift Range
Year	0.1364	1.3968	0.7087	1.2604
Month	0.0436	0.3387	0.1454	0.2951
Weekday	0.2479	0.3855	0.3141	0.1376
Traffic_Signal	0.0970	0.2251	0.1597	0.1281
Start_Lng	0.1546	0.2709	0.2084	0.1163
Start_Lat	0.1396	0.2466	0.1992	0.1070

Table 3: Feature drift analysis showing minimum, maximum, mean importance, and drift range. Year exhibits the largest drift range of 1.26.

4. Key Findings

Feature Importance Patterns

- Temporal features dominated prediction importance, with Year as the strongest predictor (1.38 SHAP value)
- Geographic features (Start_Lat, Start_Lng) consistently ranked in the top 5 across all years
- Traffic infrastructure variables (Crossing, Traffic_Signal) showed moderate importance but varied over time
- Environmental features (temperature, humidity, pressure) had surprisingly low predictive importance

Temporal Dynamics and 2020 Pandemic Effect

- The year 2020 marked a clear inflection point in feature importance patterns
- Month importance increased 7.8x from 0.044 (2019) to 0.339 (2020), indicating strong seasonality changes
- Night_Driving importance tripled during pandemic (0.044 → 0.135), reflecting changed commuting patterns
- Year feature importance surged from 0.19 (2019) to 0.88 (2020), stabilizing thereafter at ~1.40

Stable vs. Unstable Features

Stable Features (drift range < 0.05):

- Humidity, Temperature, Give_Way, Traffic_Calming, Railway
- These features maintained consistent predictive importance throughout the period

Unstable Features (drift range > 0.10):

- Year (1.26), Month (0.30), Weekday (0.14), Traffic_Signal (0.13), Start_Lng (0.12), Start_Lat (0.11)
- These features exhibited substantial importance fluctuations, particularly during transition periods

Model Insights

- Ensemble methods captured nonlinear relationships better than baseline models
- Class imbalance remained the primary performance bottleneck, limiting recall scores
- LightGBM's superiority suggests the importance of gradient-based feature interactions in accident prediction

5. Conclusions and Future Work

Research Contributions

This project made several important contributions to explainable AI in transportation safety:

- Developed a leakage-safe temporal ML pipeline with proper train-test splitting strategies
- Integrated SHAP-based explainability with temporal drift analysis
- Demonstrated that temporal analysis reveals evolving accident patterns beyond static importance
- Compared multiple ensemble boosting models for accident severity prediction
- Identified pandemic-related shifts in accident risk factors

Main Conclusions

- Temporal and geographic features are the dominant drivers of accident severity predictions
- Feature importance is not static—significant shifts occur in response to external events (e.g., COVID-19 pandemic)
- SHAP-based temporal analysis effectively captures evolving accident-risk patterns
- The 2020 pandemic represents a natural experiment revealing how traffic behavior changes impact accident dynamics

Limitations

- Binary classification (severe/non-severe) simplifies real accident severity spectrum
- No causal inference framework—analysis identifies predictive associations, not causation
- Severe class imbalance (approximately 10:1 ratio) limits minority class prediction
- Geographic features reflect location-based confounders rather than direct risk factors

Future Directions

- Extend to multiclass severity prediction (injury severity levels)
- Integrate causal inference methods (causal forests, doubly robust estimation)
- Develop spatial-temporal explainability methods
- Deploy as real-time XAI dashboard for traffic safety applications
- Apply dynamic drift detection systems for continuous model monitoring

Final Remarks

This research demonstrates the value of combining modern machine learning with explainability techniques and temporal analysis. By tracking how feature importance evolves over time, we gain insights into changing accident dynamics that static models would miss. The dramatic shifts observed around 2020 highlight how major societal events reshape traffic safety patterns, with potential implications for policy, infrastructure investment, and transportation planning.